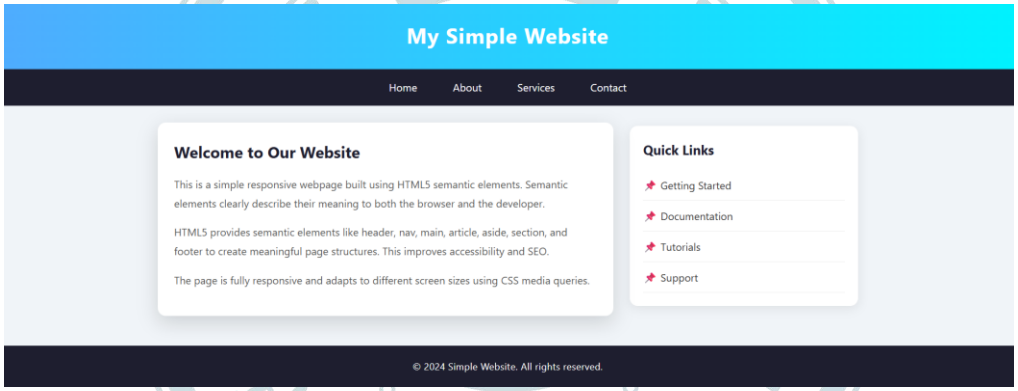
Program No: 1	Date: 5-12-2025
Program Title: Design a responsive webpage using HTML5 semantic elements and CSS.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 5-12-2025
-->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple Responsive Page</title>
 ]  <link rel="stylesheet" href="1.css">
</head>
<body>
  <header>
    <h1>My Simple Website</h1>
  </header>
  <nav>
    <ul>
      <li><a href="#home">Home</a></li>
      <li><a href="#about">About</a></li>
      <li><a href="#services">Services</a></li>
      <li><a href="#contact">Contact</a></li>
    </ul>
  </nav>
  <main>
    <article>
      <h2>Welcome to Our Website</h2>
        <p>This is a simple responsive webpage built using HTML5 semantic elements. Semantic elements clearly describe their meaning to both the browser and the developer.</p>
        <p>HTML5 provides semantic elements like header, nav, main, article, aside, section, and footer to create meaningful page structures. This improves accessibility and SEO.</p>
      <p>The page is fully responsive and adapts to different screen sizes using CSS media queries.</p>
    </article>
    <aside>
```

```
        <h3>Quick Links</h3>
        <ul>
          <li>📌 Getting Started</li>
          <li>📌 Documentation</li>
          <li>📌 Tutorials</li>
          <li>📌 Support</li>
        </ul>
      </aside>
    </main>
    <footer>
      <p>&copy; 2024 Simple Website. All rights reserved.</p>
    </footer>
</body>
</html>
```

Output

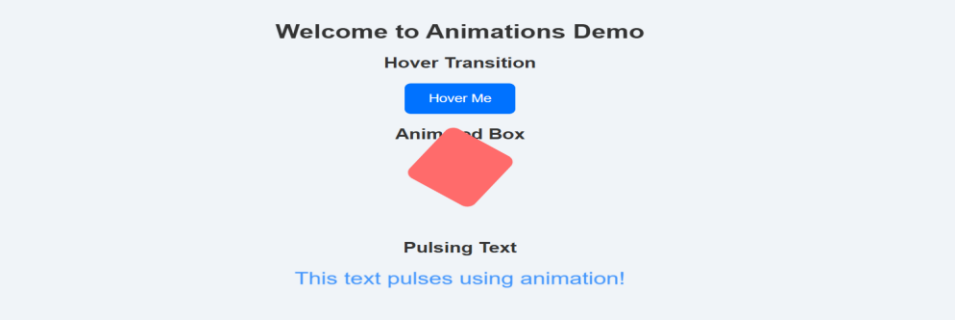


Program No: 2	Date: 5-12-2025
Program Title: Implement CSS animations and transitions on webpage elements	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 5-12-2025
-->

<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="2.css">
</head>
<body>
<header>
  <h1 class="fade-in">Welcome to Animations Demo</h1>
</header>
<section class="demo-section">
  <h2>Hover Transition</h2>
  <button class="btn">Hover Me</button>
  <h2>Animated Box</h2>
  <div class="animated-box"></div>
  <h2>Pulsing Text</h2>
  <p class="pulse-text">This text pulses using animation!</p>
</section>
</body>
</html>
```

Output



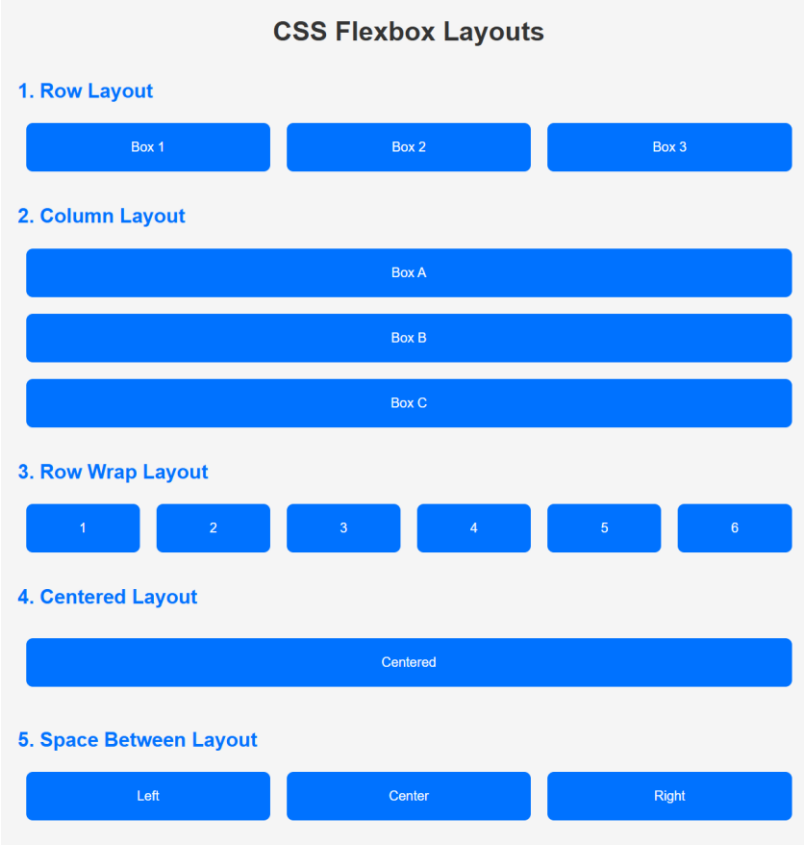
Program No: 3	Date: 5-12-2025
Program Title: Create a webpage demonstrating CSS Flexbox layouts.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 6-12-2025
-->

<!DOCTYPE html>
<html>
<head>
  <title>Flexbox Demo</title>
  <link rel="stylesheet" href="3.css">
</head>
<body>
<header>
  <h1>CSS Flexbox Layouts</h1>
</header>
<section class="flex-demo">
  <h2>1. Row Layout</h2>
  <div class="flex-row">
    <div class="box">Box 1</div>
    <div class="box">Box 2</div>
    <div class="box">Box 3</div>
  </div>
  <h2>2. Column Layout</h2>
  <div class="flex-column">
    <div class="box">Box A</div>
    <div class="box">Box B</div>
    <div class="box">Box C</div>
  </div>
  <h2>3. Row Wrap Layout</h2>
  <div class="flex-wrap">
    <div class="box">1</div>
    <div class="box">2</div>
    <div class="box">3</div>
    <div class="box">4</div>
    <div class="box">5</div>
    <div class="box">6</div>
  </div>
```

```
  <h2>4. Centered Layout</h2>
  <div class="flex-center">
    <div class="box">Centered</div>
  </div>
  <h2>5. Space Between Layout</h2>
  <div class="flex-space">
    <div class="box">Left</div>
    <div class="box">Center</div>
    <div class="box">Right</div>
  </div>
</section>
</body>
</html>
```

Output



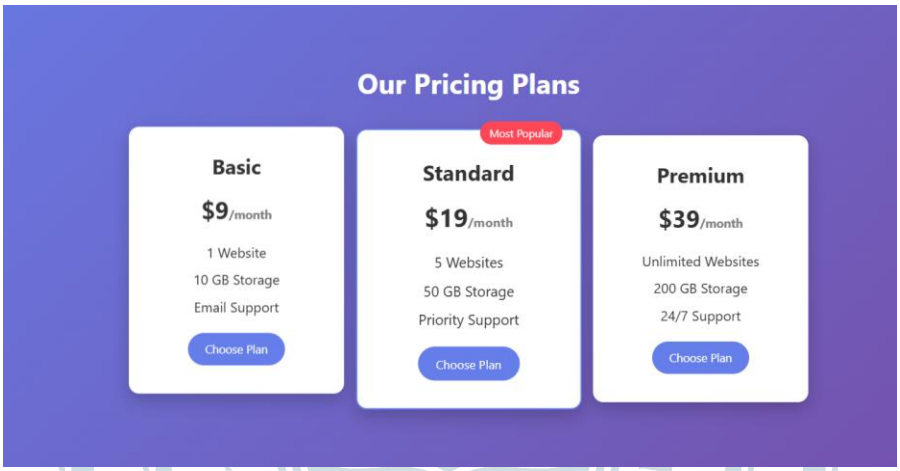
Program No: 4	Date: 5-12-2025
Program Title: Design a pricing table layout using HTML and CSS.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 6-12-2025
-->

<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="4.css">
</head>
<body>
<section>
  <h1>Pricing Plans</h1
  <div>
    <div>
      <h2>Basic</h2>
      <p>$9/month</p>
      <ul>
        <li>1 Website</li>
        <li>10 GB Storage</li>
        <li>Email Support</li>
      </ul>
      <button>Choose Plan</button>
    </div>
    <div>
      <h2>Standard</h2>
      <p>$19/month</p>
      <ul>
        <li>5 Websites</li>
        <li>50 GB Storage</li>
        <li>Priority Support</li>
      </ul>
      <button>Choose Plan</button>
    </div>
    <div>
      <h2>Premium</h2>
      <p>$39/month</p>
      <ul>
```

```
            <li>Unlimited Websites</li>
            <li>200 GB Storage</li>
            <li>24/7 Support</li>
          </ul>
          <button>Choose Plan</button>
        </div>
    </div>
</section>
</body>
</html>
```

Output



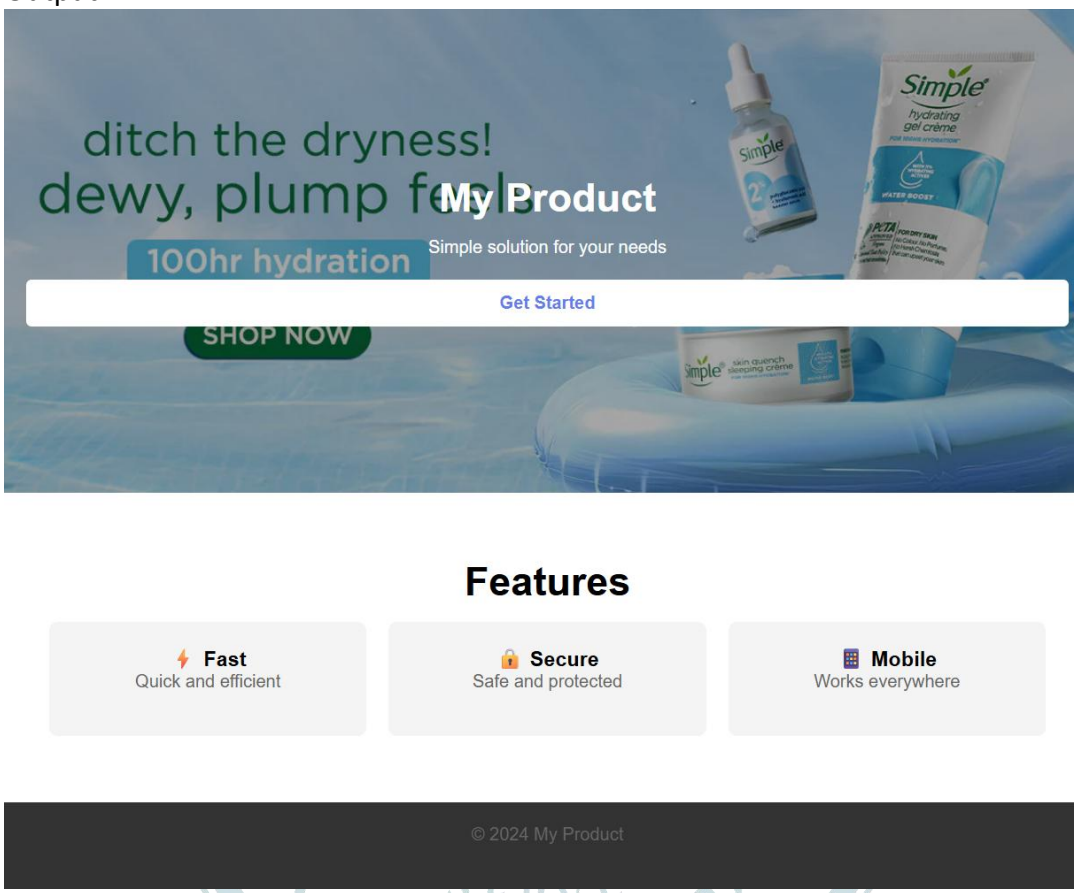
Program No: 5	Date: 12-12-2025
Program Title: Design a multi-section landing page using HTML and CSS.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 12-12-2025
-->

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Mini Landing Page</title>
</head>
<body>
  <!-- Hero -->
  <section class="hero">
    <h1>My Product</h1>
    <p>Simple solution for your needs</p>
    <a href="#" class="btn">Get Started</a>
  </section>
  <section>
    <h1>Features</h1>
    <div class="features">
      <div class="box">
        <h3> ⚡ Fast</h3>
        <p>Quick and efficient</p>
      </div>
      <div class="box">
        <h3> 🔒 Secure</h3>
        <p>Safe and protected</p>
      </div>
      <div class="box">
        <h3> 📱 Mobile</h3>
        <p>Works everywhere</p>
      </div>
    </div>
  </section>
  <footer>
```

```
    <p>&copy; 2024 My Product</p>
  </footer>
</body>
</html>
```

Output



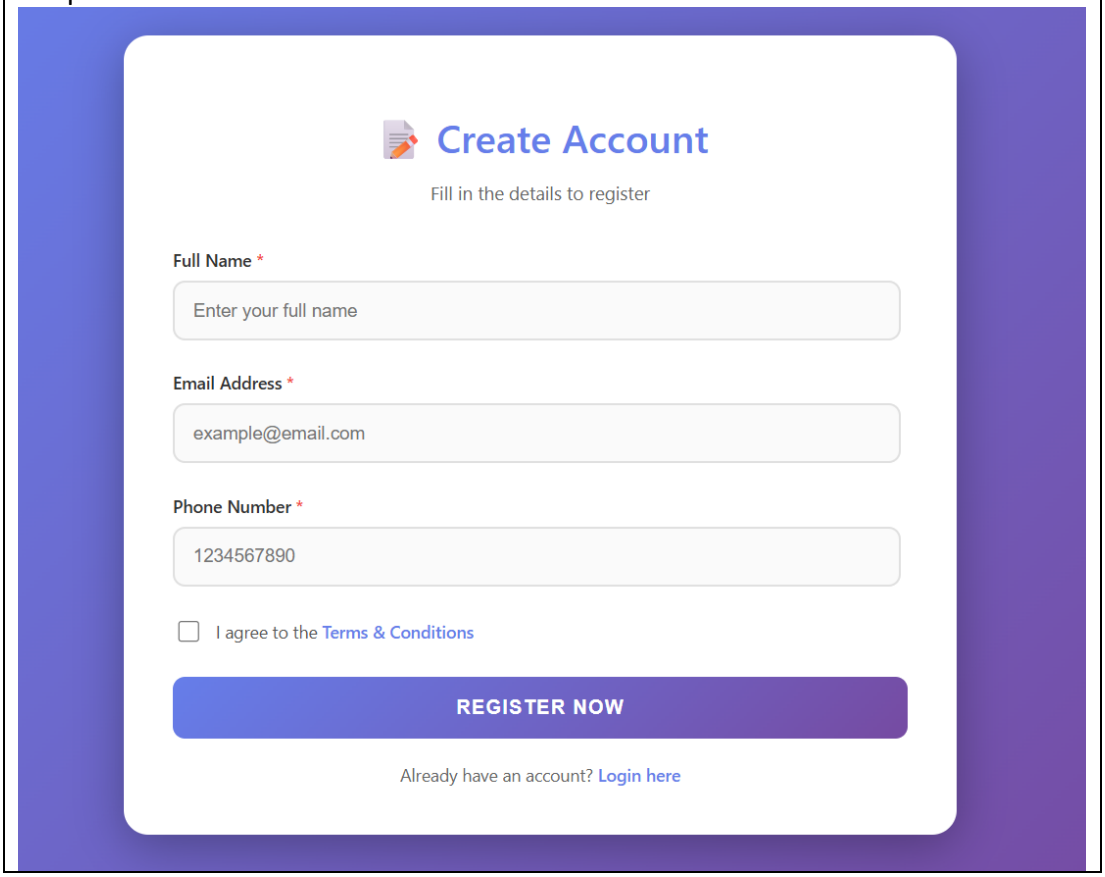
Program No: 6	Date: 12-12-2025
Program Title: Write a JavaScript program to validate a registration form	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 12-12-2025
-->

<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="6.css">
</head>
<body>
<div class="form-container">
  <h1>Create Account</h1>
  <form id="registrationForm">
    <input type="text" id="fullName" placeholder="Full Name"><br><br>
    <input type="email" id="email" placeholder="Email"><br><br>
    <input type="tel" id="phone" placeholder="Phone (10 digits)"><br><br>
    <label>
      <input type="checkbox" id="terms"> I agree to Terms
    </label><br><br>
    <button type="submit">Register</button>
  </form>
</div>
<script>
const form = document.getElementById("registrationForm");
form.addEventListener("submit", function(e) {
  e.preventDefault();
  const fullName = document.getElementById("fullName").value.trim();
  const email = document.getElementById("email").value.trim();
  const phone = document.getElementById("phone").value.trim();
  const terms = document.getElementById("terms").checked;
  if (fullName === "" || email === "" || phone === "" ) {
    alert("All fields are required!");
    return;
  }
  if (phone.length !== 10) {
    alert("Phone must be 10 digits");
    return;
```

```
    }
    if (!terms) {
        alert("Please accept Terms");
        return;
    }
    alert("Registration Successful!");
    form.reset();
});
</script>
</body>
</html>
```

Output



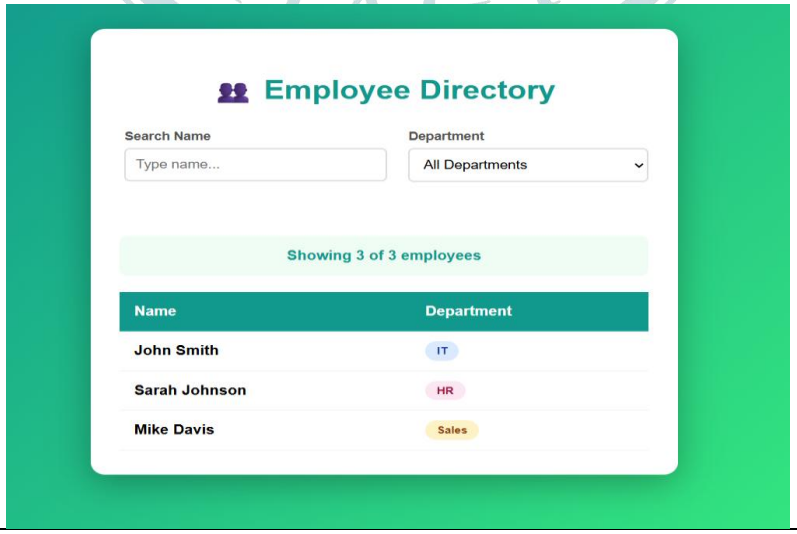
Program No: 7	Date: 12-12-2025
Program Title: Write a JavaScript program to filter and search data dynamically.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 13-12-2025
-->

<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="7.css">
</head>
<body>
<div class="container">
  <h1>Employee Directory</h1>
  <input type="text" id="nameSearch" placeholder="Search name..." oninput="filterData()">
  <select id="deptFilter" onchange="filterData()">
    <option value="">All Departments</option>
    <option value="IT">IT</option>
    <option value="HR">HR</option>
  </select>
  <p>
    Showing <span id="resultCount">0</span> of
    <span id="totalCount">0</span> employees
  </p>
  <table border="1">
    <thead>
      <tr>
        <th>Name</th>
        <th>Department</th>
      </tr>
    </thead>
    <tbody id="tableBody"></tbody>
  </table>
</div>
<script>
const employees = [{ name: "John Smith", dept: "IT", position: "Developer", email: "john@company.com" }, { name: "Sarah Johnson", dept: "HR", position: "HR Manager", email: "sarah@company.com" },
```

```
  { name: "Mike Davis", dept: "Sales", position: "Sales Rep", email: "mike@company.com" }];
function displayData(data) {
  const tbody = document.getElementById("tableBody");
  document.getElementById("resultCount").textContent = data.length;
  document.getElementById("totalCount").textContent = employees.length;
  if (data.length === 0) {
    tbody.innerHTML = "<tr><td colspan='4'>No employees found</td></tr>";
    return;
  }
  tbody.innerHTML = data.map(emp => `
    <tr>
      <td>${emp.name}</td>
      <td>${emp.dept}</td>
    </tr>
  `).join('');
}
function filterData() {
  const name
 = document.getElementById("nameSearch").value.toLowerCase();
  const dept = document.getElementById("deptFilter").value;
  const filtered = employees.filter(emp =>
    emp.name.toLowerCase().includes(name) &&
    (!dept || emp.dept === dept)
  );
  displayData(filtered);
}
function resetFilters() {
  document.getElementById("nameSearch").value = ""
```

Output



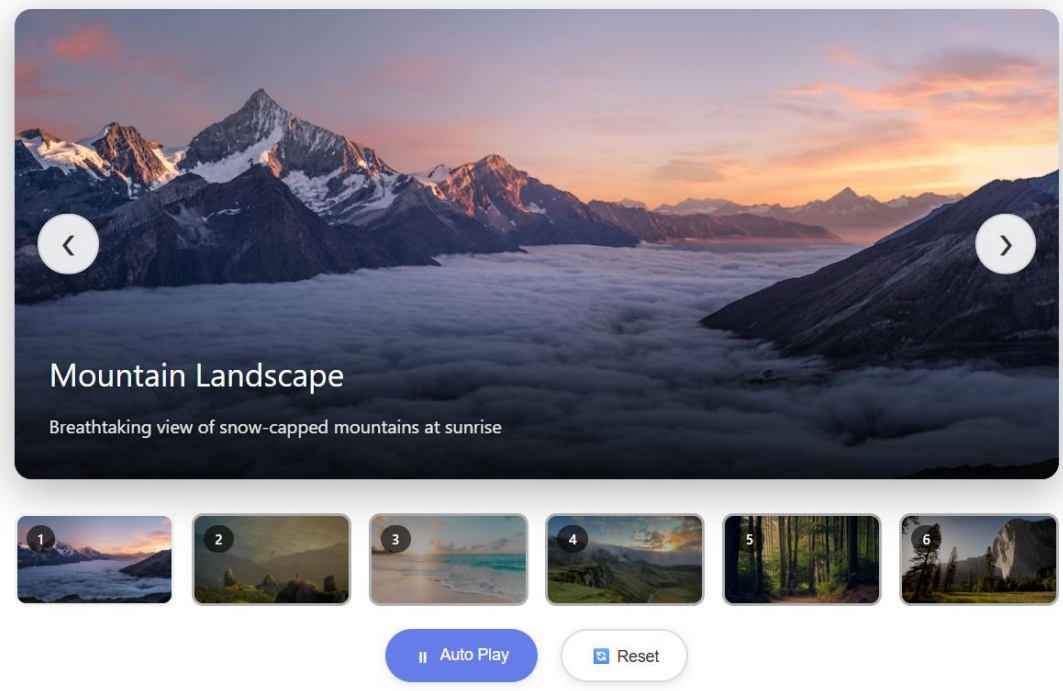
Program No: 8	Date: 12-12-2025
Program Title: Implement image slideshow using JavaScript and CSS Grid.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 13-12-2025
-->

<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="8.css">
</head>
<body>
<div class="slideshow-container">
  <div class="slide">
    <img id="slideImage" src="">
    <h2 id="title"></h2>
    <p id="description"></p>
  </div
  <button onclick="changeSlide(-1)">❮ Prev</button>
  <button onclick="changeSlide(1)">Next ❯</button>
</div>
<script>
const images = [
  {
    url: "https://images.unsplash.com/photo-150695621bda4d3df4?w=800”,
    title: "Mountain",
    description: "Beautiful mountain view"
  },
  {
    url: "https://images.unsplash.com/photo-150695621bda4d3df4?w=800”,
    title: "Forest",
    description: "Green forest pathway"
  },
  {
    url: "https://images.unsplash.com/photo-150695621bda4d3df4?w=800”,
    title: "Ocean",
    description: "Peaceful ocean scene"
  }
```

```
];
let currentSlide = 0;
function showSlide(index) {
   document.getElementById("slideImage").src = images[index].url;
   document.getElementById("title").textContent = images[index].title;
   document.getElementById("description").textContent                    =
images[index].description;
}
function changeSlide(direction) {
   currentSlide = (currentSlide + direction + images.length) % images.length;
   showSlide(currentSlide);
}
showSlide(currentSlide);
</script>
</body>
</html>
```

Output

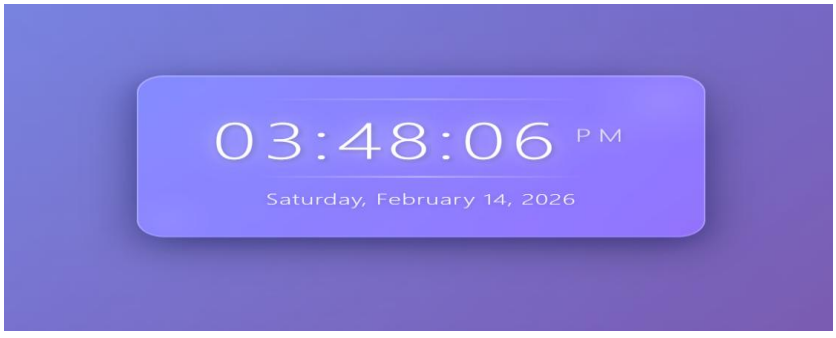


Program No: 9	Date: 19-12-2025
Program Title: Create a real-time digital clock using JavaScript.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 19-12-2025
-->

<!DOCTYPE html>
<html>
<head>
   <link rel="stylesheet" href="9.css">
</head>
<body>
<div class="clock-container">
   <div id="time">00:00:00</div>
   <div id="date"></div>
</div>
<script>
function updateClock() {
   const now = new Date();
   const time = now.toLocaleTimeString();
   const date = now.toDateString();
   document.getElementById("time").textContent = time;
   document.getElementById("date").textContent = date;}
updateClock();
setInterval(updateClock, 1000);
</script>
</body>
</html>
```

Output



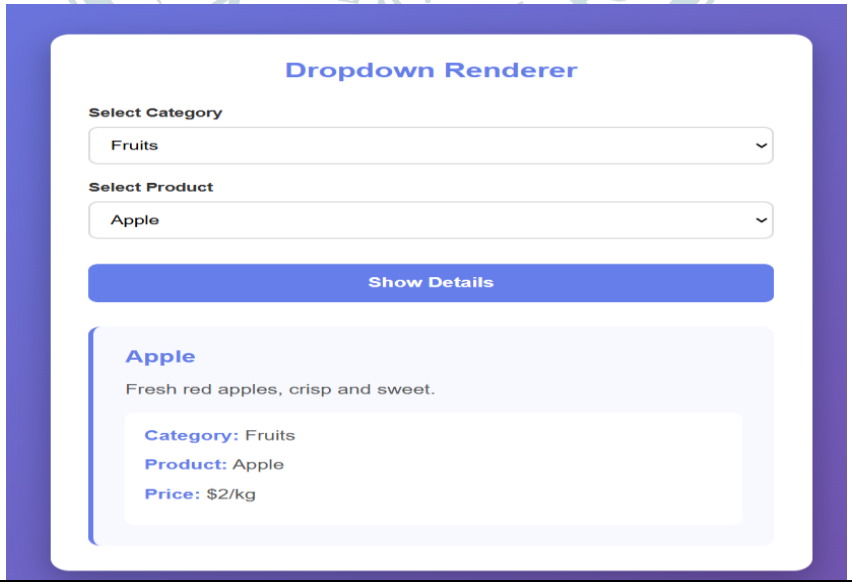
Program No: 10	Date: 19-12-2025
Program Title: Implement dropdown-based content rendering using JavaScript	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 19-12-2025
-->

<!DOCTYPE html>
<html>
<head>
  <title>Dropdown Renderer</title>
  <link rel="stylesheet" href="10.css">
</head>
<body>
<div class="container">
  <h1>Dropdown Renderer</h1
  <label>Select Category</label>
  <select id="category" onchange="updateProducts()">
    <option value="">-- Choose Category --</option>
    <option value="fruits">Fruits</option>
    <option value="vegetables">Vegetables</option>
  </select>
  <label>Select Product</label>
  <select id="product">
    <option value="">-- Choose Product --</option>
  </select>
  <button onclick="renderContent()">Show Details</button>
  <div id="contentBox">
    <h2 id="title"></h2>
    <p id="description"></p>
    <p><strong>Category:</strong> <span id="categoryName"></span></p>
    <p><strong>Price:</strong> <span id="price"></span></p>
  </div>
</div>
<script>
const data = { fruits: {
    apple: { name: "Apple", price: "$2/kg", desc: "Fresh red apples." },
    banana: { name: "Banana", price: "$1.5/kg", desc: "Sweet yellow bananas."
} },
  vegetables: {
```

```
      tomato: { name: "Tomato", price: "$2.5/kg", desc: "Fresh tomatoes." },
      carrot: { name: "Carrot", price: "$1.8/kg", desc: "Crunchy carrots." }
   } };
function updateProducts() {
   const category = document.getElementById("category").value;
   const product = document.getElementById("product");
   product.innerHTML = '<option value="">-- Choose Product --</option>';
   if (data[category]) {
      for (let key in data[category]) {
         product.innerHTML += `<option
 value="${key}">${data[category][key].name}</option>`;
      } } }
function renderContent() {
   const category = document.getElementById("category").value;
   const product = document.getElementById("product").value;
   if (!category || !product) {
      alert("Select category and product");
      return;
   }
   const item = data[category][product];
   document.getElementById("title").textContent = item.name;
   document.getElementById("description").textContent = item.desc;
   document.getElementById("categoryName").textContent = category;
   document.getElementById("price").textContent = item.price;
}
</script>
</body>
</html>
```

Output



Program No: 11	Date: 19-12-2025
Program Title: Write a program to display student details (name, age, grade) using JSON.	

```
<!--
@ Rosha Thankachan
Roll no:50
Date: 20-12-2025
-->

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Details</title>
  <link rel="stylesheet" href="11.css">
</head>
<body>
  <div class="container">
    <div class="header">
      <h1>Student Management System</h1>
    </div>

    <div class="stats">
      <div class="stat-item">
        <div class="stat-value" id="totalStudents">0</div>
        <div class="stat-label">Total Students</div>
      </div>
      <div class="stat-item">
        <div class="stat-value" id="avgAge">0</div>
        <div class="stat-label">Average Age</div>
      </div>
    </div>
    <div class="controls">
          <input type="text" id="searchInput" placeholder="Search by name..."
onkeyup="filterStudents()">
      <select id="gradeFilter" onchange="filterStudents()">
        <option value="">All Grades</option>
        <option value="A">Grade A</option>
        <option value="B">Grade B</option>
      </select>
```

```
        <button onclick="showAllStudents()">Show All</button>
      </div>
      <div class="student-grid" id="studentGrid"></div>
   </div>
<script>
const studentsData = [
   { id: 1, name: "Alice Johnson", age: 20, grade: "A" },
   { id: 2, name: "Bob Smith", age: 22, grade: "B" },
   { id: 3, name: "Charlie Brown", age: 21, grade: "A" }
];
let currentStudents = [...studentsData];
function init() {
   displayStudents(currentStudents);
   updateStatistics(currentStudents);
}
function displayStudents(students) {
   const grid = document.getElementById('studentGrid');
   if (students.length === 0) {
      grid.innerHTML = "<p>No Students Found</p>";
      return;
   }
   grid.innerHTML = students.map(student => `
      <div onclick="showDetails(${student.id})">
         <h3>${student.name}</h3>
         <p>Age: ${student.age}</p>
         <p>Grade: ${student.grade}</p>
      </div>
   `).join('');
}
function updateStatistics(students) {
   document.getElementById('totalStudents').textContent = students.length;
   if (students.length === 0) {
      document.getElementById('avgAge').textContent = 0;
      return;
   }
}
function filterStudents() {
   const search = document.getElementById('searchInput').value.toLowerCase();
   const grade = document.getElementById('gradeFilter').value;
   currentStudents = studentsData.filter(s =>
      s.name.toLowerCase().includes(search) &&
      (!grade || s.grade === grade)
   );
   displayStudents(currentStudents);
```

```
    updateStatistics(currentStudents);
}
function showAllStudents() {
    document.getElementById('searchInput').value = '';
    document.getElementById('gradeFilter').value = '';
    currentStudents = [...studentsData];
    displayStudents(currentStudents);
    updateStatistics(currentStudents);
}
function showDetails(id) {
    const student = studentsData.find(s => s.id === id);
    if (student) {
        alert(
`Student Details
Name: ${student.name}
Age: ${student.age}
Grade: ${student.grade}
ID: ${student.id}`
        );
    }
}
window.addEventListener('load', init);
</script>
</body>
</html>
```

Output

